

Reading 6.1

Started: Apr 22 at 4:17pm

Quiz Instructions

Question 1

8 pts

Using the following grammar from Chapter 6, arrange the sentential forms for deriving the string "the student deserves the vegetable".

<sentence> → <noun-phrase> <verb-phrase>

<noun-phrase> → the <noun>

<verb-phrase> → <verb> <noun-phrase>

<verb> → moves | annoys | deserves

<noun> → student | cow | vegetable | worm | pot

1

[Choose]



2

[Choose]



3

[Choose]



4

[Choose]



5

[Choose]



6

[Choose]



7

[Choose]



8

[Choose]

**Question 2****4 pts**

Complete the derivation of $aaaabb$ using the grammar in Example 6-3 (repeated below).

 $S \rightarrow aSb \mid X \mid Y$ $X \rightarrow aX \mid a$ $Y \rightarrow Yb \mid b$

$S \Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow aaaabb$

The following CFG from Example 6–5 applies to the next 2 questions.

$S \rightarrow aSa \mid bSb \mid X$ $X \rightarrow aYb \mid bYa$ $Y \rightarrow aY \mid bY \mid \lambda$ **Question 3****6 pts**Complete the derivation of the string *abaabb*.

$S \Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow$

$\Rightarrow$ $\Rightarrow$ *abaabb*

Question 4**4 pts**Complete the derivation of the string *bbbabb*.

$S \Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow$ *bbbabb*

Question 5**5 pts**

Complete the derivation of the string *aabba* using the grammar in Example 6-6:

$$S \rightarrow XY$$
$$X \rightarrow aXb \mid \lambda$$
$$Y \rightarrow aY \mid \lambda$$

$S \Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow$
 $\Rightarrow aabba$

The next two questions use the following grammar from Example 6-9:

$$S \rightarrow SS \mid aSb \mid bSa \mid \lambda$$

Question 6

3 pts

Complete the derivation of the string *ababab*.

$S \Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow ababab$

Question 7**7 pts**

Complete the derivation of the string *ababbaabba*.

$S \Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow$ $\Rightarrow$
 $\Rightarrow$ $\Rightarrow$ $\Rightarrow ababbaabba$

The next 2 questions pertain to the grammar below:

$$S \rightarrow a \mid aA \mid B \mid C$$

$$A \rightarrow aB \mid \lambda$$

$$B \rightarrow Aa$$

$$C \rightarrow ccC$$

$$D \rightarrow ddd$$
Question 8**2 pts**

Which variables in the grammar above are non-terminating?

☐ A

☐ B☐ C☐ D**Question 9****2 pts**

Which variables in the grammar above are unreachable?

☐ A☐ B☐ C☐ D

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